Intern Survival Guide: Continuum Normalization & Line Fitting for OB Stars

Companion document to the SARL Internship Offer — keep this open while working.

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0. Shared Conventions (Lock These In During Week 1)

Since you and your co-intern are building on the same pipeline for your own 50-star subsets, agree on these column names and values together before either of you writes fitting code. Mismatched conventions are the most common cause of a messy Week 6 merge.

Column	Purpose	Example values
`star_id`	Unique identifier per star, consistent with the shared archive naming	e.g. SDSS plate-MJD-fiber
`analyst`	Who processed this star	`"Student A"`, `"Student B"` (use your actual names if you prefer — just agree on the format)
`quality_flag`	Fit/data quality status	`"ok"`, `"low_SNR"`, `"fit_failed"`, `"integration_fallback"`

Use exactly these strings (or an agreed alternative) — a validation script comparing your two tables will assume matching values. If you invent a new flag mid-project (e.g. for a new edge case you hit), tell your co-intern immediately so both of you use it consistently.

Week 2 sync: Before scaling past your first handful of stars, compare your continuum normalization approach and a few fit outputs with your co-intern. Catching a systematic difference (e.g. a different line-free window or clipping threshold) here is much cheaper than catching it during the Week 6 merge.

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1. Continuum Normalization — Do's and Don'ts

Goal: Divide the observed spectrum by a smoothly varying continuum so that absorption/emission lines sit near y = 1.

Pre-defined line-free windows (use these):

Wavelength range (Å)	Purpose
4050 – 4080	Blue anchor

Wavelength range (Å)	Purpose
4400 - 4450	Avoids Hγ and He lines
5900 - 6000	Clean region for B stars
6650 - 6750	Red side of Hα

Do:

- Fit a low-order polynomial (degree 2–4) or a spline with few knots (e.g., `scipy.interpolate.UnivariateSpline` with `s=0.1*len(flux)`)
- Iteratively clip outliers (>3σ) to suppress residual line wings
- Store the continuum value **at each line center** separately — this is your denominator in the EW formula

Don't:

- Extrapolate splines beyond the last line-free window — clip to the wavelength range of the data
- Use a high-order polynomial — it overfits noise and introduces false wiggles
- Normalize each line individually — this introduces artificial offsets between lines

Quick sanity check: After normalization, the mean flux in line-free regions should be 1.0 ± 0.05. If not, revisit your continuum windows.

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2. Line Fitting — Common Edge Cases

Case A: Blended lines (e.g., Hγ + nearby Fe I)

- Fit a **multi-component model** (e.g., two Gaussians + linear baseline). `pymultifit` supports this natively.
- Failsafe: if the secondary component amplitude is < 2× the noise level, drop it and refit with a single component.

Case B: Very broad Balmer lines (rotation or pressure broadening)

- Use a **Lorentzian** profile — it handles extended wings better than a Gaussian.
- Fix the continuum level to the previously stored value to reduce free parameters.

Case C: Low SNR (SNR < 10 per pixel)

- Do not trust EW measurements from these spectra.
- Flag: `quality_flag = "low_SNR"` and exclude from population statistics.

- If you must attempt a fit, fix the line width to a physically motivated prior (e.g., $\sigma = 2$ Å for Hα in mid-B stars).

Case D: Bad pixels / cosmic rays

- Detect outliers as >5σ from the local median in a 5-pixel sliding window.
- Replace flagged pixels with linearly interpolated values before fitting.
- `astropy.nddata` utilities can help here.

Case E: Emission lines (Be candidates)

- For EW(Hα) < 0, **use a Gaussian** rather than Lorentzian — emission profiles in early-type Be stars are generally more symmetric.
- Always visually inspect the 10 most negative EW(Hα) measurements before reporting them as candidates — continuum placement artifacts can mimic weak emission.

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3. Quick QA Checklist (run before processing your full 50-star subset)

You and your co-intern will both work through this checklist against the same golden set in Week 1 — treat it as a joint learning exercise the first time through, not just a gate. For **each** of the golden validation stars provided by the supervisor, verify:

Check	Acceptable range
Continuum level at line center	0.95 – 1.05
Reduced χ^2 of fit	< 2.0
EW(Hα) vs. literature value	within 20% or 0.5 Å (whichever is larger)
EW(Hβ) < EW(Hα) in absorption	yes (Balmer decrement expected)
He II (4686 Å) detected only in O stars	consistent with supervisor-provided classification

> **Note on He I lines:** He I 4471 Å and He I 5876 Å do **not** have equal EW in general — their ratio varies across B subtypes due to differing excitation conditions. Do not use a fixed ratio as a validation criterion. Instead, compare your measured EWs for each line separately against the literature values in the golden set.

If **any** validation star fails, debug your normalization or fit model **before** scaling up to your 50-star subset. It's normal to iterate on this a few times — the golden set exists precisely so mistakes get caught here rather than downstream.

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4. What to Do When a Fit Fails

```python
def safe_fit(spectrum, line_center, model='gaussian'):
    try:
        fit = fit_line(spectrum, line_center, model)
        if fit.chi2_red > 3.0 or fit.amplitude_err > abs(fit.amplitude):
            raise RuntimeError("Poor fit quality")
        return fit
    except Exception:
        # Fallback: measure EW by direct trapezoidal integration
        # within line_center ± 10 Å (adjust window per line)
        ew_direct = integrate_region(spectrum, line_center - 10, line_center +
        return FailedFit(ew_direct, quality_flag='integration_fallback')
```

The fallback integration EW is less precise but recovers a measurement rather than losing the spectrum entirely. Always record the `quality_flag` in your results table — it matters for downstream statistics.

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5. Be Star Candidate Threshold — Why −5 Å?

The classical Be star identification criterion in the literature (Jaschek & Jaschek, and subsequent surveys) uses **EW(Hα) < −5 Å** as the minimum emission threshold. This is deliberately conservative: it excludes noise-driven false positives at typical SDSS spectral SNR, and ensures candidates show unambiguous circumstellar emission rather than marginal continuum artifacts.

In this internship, the two-gate criterion is:

1. EW(Hα) < −5 Å (amplitude gate)
2. |EW(Hα)| / σ_{EW} > 3 (significance gate)

Both must be satisfied. Candidates passing only the amplitude gate but not the significance gate are logged separately as **marginal** — they are worth visual inspection but should not be reported as confirmed candidates.

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6. Before the Week 6 Merge

A few practical steps that make combining your 50-star table with your co-intern's painless:

- Tag every row with `analyst = "Student A"` (or your agreed name) from the start, not retroactively — it's easy to forget and hard to reconstruct later

- Run a lightweight code review on each other's fitting scripts at least once before merging — a second pair of eyes tends to catch silent unit or sign-convention mistakes that a chi-squared check won't
- Diff your `quality_flag` value counts against each other's (e.g. how many `"low_SNR"` per subset) — a large discrepancy usually means one of you has a stricter or looser SNR cut than intended, not that one subset is genuinely worse
- Re-run the Section 3 QA checklist on a handful of your co-intern's stars (and vice versa) as a final cross-check before the combined catalog goes into the report

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Companion document prepared by SARL, IST